Machine Learning Prediction of Intensive Care Length of Stay for Lung Cancer Patients

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Abstract

Acknowledgements

1 Introduction

1.1 Aims and Objectives

The main aim of this project is to build accurate and efficient machine learning models using data from the MIMIC dataset to enhance the prediction of the length of stay in the Intensive Care Unit, ICU, for lung cancer patients. The ideal result would be for this to benefit personalised healthcare and resource management in oncology.

The main aims and objectives are the following:

- ~~Aim~~ 1 - Extract the data from the MIMIC dataset to form a final dataset ready to be used to develop predictive models.
  - Objective 1 - Determine the variables for selection
  - Objective 2 - Setup the MIMIC database on PostgresSQL
  - Objective 3 - Extract the data
  - Objective 4 - Clean the data
- ~~Aim~~ 2 - Train various predictive models.
  - Objective 1 - Cross-validate
  - Objective 2 - Impute the missing data
  - Objective 3 - Linear regression model
  - Objective 4 - Penalistation models
  - Objective 5 - Random forest
- ~~Aim~~ 3 - Evaluate the models assessing the effectiveness of each.
  - Objective 1 - Calculate performance metrics
  - Objective 2 - Compare models
  - Objective 3 - Compare findings with other studies

1.2 Lung Cancer Background

Cancer is a disease that involves the uncontrollable multiplication of malignant cells, in some cases forming tumours in organs or tissues. In the UK, and many other countries around the world, cancer has become the leading cause of death, and in the UK specifically it has been reported that around 1 in 2 people will be diagnosed with cancer at some point in their lives. This statistic indicates that the volume of patients requiring ICU treatment will be significantly large and therefore the optimisation of length of stay analysis is key to resource management and patient care planning.

1.3 Length of Stay Prediction

The length of stay in an ICU ward is an important piece of information for clinicians to know and if it can be predicted accurately the efficiency of the intensive care process would improve and so would the quality of care provided. At the time of writing, although some machine learning predictions of ICU length of stay are being researched and perhaps trialled in a limited amount of hospitals, the majority of healthcare systems go by clinical judgement alone. One of the main reasons behind this stems from the lack of understanding regarding how to pair the machine learning assistance to the clinician's expertise in a way that optimises the blend of predictive knowledge that both provide[30]. The lack of use of these methods in practice provides a rationale for conducting this type of study.

1.4 Machine Learning's Possible Benefits

1.5 The MIMIC Database

MIMIC is a large relational database that consists of deidentified data collated from thousands of admissions to the intensive care unit of the Beth Israel Deaconess Medical Centre in Boston, Massachusetts. The acronym MIMIC has stood for "Multiparameter Intelligent Monitoring in Intensive Care" and more recently "Medical Information Mart for Intensive Care". Some key features of this database are that it is the only freely accessible critical care database of its kind and contains detailed information on patients spanning more than a decade. The most recent version of the MIMIC database is MIMIC-IV which includes, ~~most importantly~~, data ~~all the way up to the year 2019 as opposed to 2012 for MIMIC-III~~. ~~Using the updated version should allow for greater relevance and accuracy of the prediction model.~~ The value of MIMIC in this study lies in the comprehensive real-world data it contains, which enables the model to train more accurately compared to synthetic

data, allowing it to fit better to unseen data.

Data can be accessed after completing a short ethics course set up by the CITI programme website.[citeMIMICethics]

1.6 GitHub Repository

~~A GitHub repository has been set up to back up files so that lost work can be retrieved in the event of unexpected technical issues. This will be updated throughout the course of the project.~~[cite github page]

2 Methods and Materials

2.1 Variable Selection

Variables to use in the prediction models must be determined before any data extraction from the MIMIC-IV database can occur. The importance of choosing relevant variables is significant as this has a strong impact on the performance of the model. ~~For example, a variable such as the amount of languages the patient speaks would most probably be fairly useless in predicting lung cancer length of stay in the ICU.~~ Therefore some prior knowledge on the key variables that affect lung cancer is needed, to obtain this it would be useful to draw on medical knowledge. In the absence of first hand medical knowledge some research on the variables used in previous ~~study~~ must be examined. A study with the same aim of prediction for the length of stay for lung cancer patients using the MIMIC-III database is found [21]. It is not clear that any other study focuses primarily on prediction of length of stay for lung cancer patients, but the variables included here would be those that are explicitly useful for this prediction. They mention the methods of obtaining the variables they have included is a process of first considering clinical significance and then using a process of recursive feature elimination as a way of removing the less useful variables. The variables that remained are as listed: age, temperature, platelets, admission (specifically emergency), glucose, chloride, potassium, pulse oximetry, PTT, anion gap, respiratory rate, haemoglobin, troponin, calcium gluconate, haematocrit, tidal volume, diastolic, ondansetron, INR, sodium. Of these pulse oximetry, anion gap, respiratory rate, tidal volume and diastolic were not collected due to lack of data. The variables sex, ethnicity and health insurance were also included with the reason that they may be clinically significant.

Whether a patient leaves the ICU through discharge or through death is important to consider as a prediction model that doesn't take this into account may predict those who end up dying having a much longer length of stay than they do based on the data that indicates that they may have poor health.

Age is treated as a continuous variable but is limited by a minimum of 18 to remove non-adults from the study. Temperature, platelets, glucose, chloride, potassium, INR, PTT, haemoglobin, troponin, haematocrit and sodium were also all treated as continuous variables. Sex, death in the ICU, calcium gluconate and ondansetron were binary variables and the rest treated as categorical. Chloride, sodium and potassium were given in units of mEq/L, platelets in k/uL, glucose in mg/dL, haemoglobin in g/dL, haematocrit as a percentage, PTT in seconds, troponin in ng/L, temperature in degrees celsius and age in years. Ethnicity was collected into many levels which will be collated in the data cleaning phase, the same for admission type and health insurance. Sex was split into male and female and as for the calcium gluconate and ondansetron variables they were split into two groups based on whether the patient had taken the drug or not.

The target variable, length of stay, is to be treated as a continous variable measured in hours, meaning regression modelling is more suitable than classification in this context.

2.2 Database Management

In order for data to be extracted from the MIMIC database, the data files from physionet should be downloaded and exported into a database schema in pgAdmin 4. From there the database can be queried with SQL commands for data extraction.

To create the schema, ~~this being the frame housing the tables that constitute the database~~, a SQL script is run, specifically tailored for MIMIC-IV, to construct empty tables ready for the data export process. The script was retrieved from a Github repository.[32]

After ~~running the script and having~~ the empty database ~~setup,~~ the data files will need to be prepared ready for export. Some data files were ~~quite large,~~ too large for pgAdmin to export in one process. Therefore the larger data files were split into smaller parts with the use of the Windows powershell command line.[cite command code] Once ~~all the~~ larger data files had been split they were exported into their respective tables in the schema.

2.3 Data Extraction

Now that the MIMIC database has been setup the data for the determined variables can be extracted.

To obtain the cohort from MIMIC for analysis, the ICD-10 [16]codes corresponding to the diagnosis of lung cancer are needed to identify the patients required for the study. ICD-10 codes represent various patient diagnoses in alphanumeric form, making it easier to record diagnostic data, thereby reducing the likelihood of incorrect entries. The ICD-10 codes that indicate a patient has lung cancer are the C34 codes, identifying a ”Malignant neoplasm of bronchus and lung”, or the C78.0 codes, identifying a ”Secondary malignant neoplasm of lung”. A secondary malignant neoplasm is a cancer that develops as a result of a previous cancer. Along with the ICD-10, MIMIC-IV also uses ICD-9 to indentify health conditions. The ICD-9 codes that correspond to the lung cancer cases that are to be extracted are the 162.9 codes. ICD-10 codelist is the updated version of the ICD-9. ICD-10 was introduced in the middle of the data collection process, ICD-10 was introduced in 2015 and MIMIC-IV was collected between 2008 and 2018. This is the reason both set of codes appear. ~~At the time of writing this report the most up to date ICD codelist in use is ICD-11. If MIMIC-V were to be released, it would be expected that ICD-11 codes would appear.~~

Through pgAdmin 4, SQL queries were written to sift through the database to extract the data to build a dataset containing the desired output laid out in the script. One of these scripts can be used to find data on the lung cancer patients with ICU stays by extracting from the relevant tables.

Similarly, to find data on each variable separate scripts were written each producing datasets extracted from all the data in the database. For example, a SQL query for temperature data would produce a dataset showing every temperature recording in the database. Later, these datasets were to be used to match each admission to their corresponding records, creating a dataset ready for building the machine learning models. Within the SQL scripts there is a set of commands that reduces the data down to only those patients with ICU stays. Since this study does not concern hospital stays without an ICU admission, these commands were utilised so that each of the datasets included only admissions with ICU stays. The data required is located by finding the specific label that identifies the data needed for extraction, for example, for finding temperature data, the WHERE command is used to find data labelled in part with ”temp” in the labels table.

Should there be multiple values before the first ICU admission the value recorded closest to the admission is extracted. Only data recorded within at least 1 hour of being admitted to the ICU

ward were extracted.

The datasets, now compiled, were saved into CSV files and imported into one R file. After adjustment, as outlined in the data cleaning section, the datasets should all be collated into one.

2.4 Data Cleaning

Some data cleaning was executed in the SQL queries with the rest written in the R code.

In SQL, as mentioned before, the recordings were reduced down to admissions with ICU stays only to avoid collecting excessive data that would not be of any use going forward. For admissions with multiple values before the ICU stay, the value closest to the ICU stay start time is taken with others discarded.

The study treats each ICU stay spaced out by a significant amount of time. ~~The models are supposed to be trained to predict ICU lenght of stay not hospital length of day.~~ The ICU length of stay value is defined by the length of stay of the first ICU stays and the duration of any readmissions that occur within an hour of prior discharge. An admission occuring outside a day between prior discharge and the start of admission is considered, a seperate admission in the data, even though these both come from the same patient. An extra variable is created, indicating the number of previous ICU stays.

Some of the categorical variables, extracted from the dataset, had many specific levels with some including an insufficiently low amount of individuals. For example the ethnicity variable was ~~split down into many groups~~, some caputuring only a few individuals ~~as these groups in the wider population are considered minority ethnic~~. ~~Thankfully~~ to remedy this many of the groups fall under the same ethnic umbrella, white - eastern/european and white - russian although specifically different ethnic groups, individuals from both groups are all considered white. Individuals from ethnic groups that don't fit into a specific category but still need to be categorised can be placed in an other category. The limitations of this are apparent in that the individuals in this group have a significant amount of variance in their ethnic characteristics which is not helpful when using ethnicity as a predictor in a prediction model. Also the aggregation of groups in this way reduces the validity of the true impact of ethnicity on a patient's length of stay. It is important to aggregate the categories in this way for personal security purposes, if a patient is known to be the only patient in a certain category with a lung cancer intensive care stay they would be easily identifiable. In a similar way the extracted admission type groups were also aggregated.

Any missing data in a categorical variable was placed into its own missing data group.

2.5 Missing Values

Missing values in the data act as a roadblock to meaningful analysis. Many machine learning models require the input data to be complete for them to work, and their performance is greatly improved when the missingness is remedied so that the resulting study population matches the true population distribution as closely as possible[19].

There are three distinct types of missing data, MAR (missing at random), MNAR (missing not at random) and; MCAR (missing completely at random). Data is considered MCAR when the missingness of the data is ~~not reliant on external factors whether that be those recorded within the study or those without~~. This differs from MNAR which describes missing data that depends solely on data outside the study. And finally MAR is dependent on the data within the study[29]. The importance of these distinctions lie in the determination of the choice of imputation or deletion method of dealing with this missingness.

Dealing with missing data can involve the complete case analysis omission of records or imputation. Although omission is the simpler method, the resulting model is subject to bias as there due to the possibility for certain groups to have missing data, of the type MAR or MNAR, in some variables. Therefore, omission of these data could potentially lead to a misrepresentative study population which in turn leads to the aforementioned bias in the prediction models[26].

A much more valid approach, to minimise bias, is imputation. Imputation is, in simplistic terms, the filling in of missing data, as an approximation, to provide a full dataset for analysis, The imputed data is based on the statistical characteristics of the observed data such as its distribution. For example, if the a variable's distribution is normal, the imputed data should lie somewhere on the same normal distribution with the imputation method mapping the data in this way.

There are a few methods of data imputation. Single imputation is a basic imputation method that imputes the missing data with single estimated values. One way to singularly impute the data is to use a statistic such as the mean or the median for numerical data, or the mode for categorical[14]. For datasets with a significant amount of missing data this imputation method is less suitable and not recommend for producing robust estimates of the missing data as the variance of the resulting imputed distribution would almost certainly be a poor estimation of the variance of the true distribution. Other methods offer an improvement to this simplistic form of imputation. The hot deck method matches individuals based on characteristics, such as age and sex, with which they share sufficient similarity[4] and uses these groups to impute the missing data in the dataset.

The individual with the missing data is labelled the recipient and the individual with the data used to impute the missing value is labelled the donor. How the donor is selected is dictated by the type of hot deck imputation used [28]. of imputing all the missing data with the same value, methods such as hot deck or regression give imputations that differ depending on the characteristics of each record. Regression models for imputation also use data from other variables by using established regression techniques such as those used to predict the length of stay in this study, for example linear regression and random forest. It is worth noting that both these methods require the external variables used to be complete, where you have datasets with a lot of missing data these methods are less effective. Despite the improvements from using these methods, single imputation is not the optimal method when missingness in the dataset is more prevalent and therefore advanced methods should be considered.

Multiple imputation improves on single imputation in severak ways with the use of multiple approximations for the missing data and allowing for every variable, including those with missing data, to be used in the imputation process[5]. MI works without modification if the missing data included in the dataset are either MAR or MCAR. For data that is MNAR sometimes with the appropriate adjustments to the data multiple imputation methods are suitable.

A common MI method is known as multivariate imputation by chained equations (MICE) (2). This method imputes every variable using data from the other variables as predictive inputs the defined model whether it be regression, random forest or some other kind.

The MICE method involves an iteration process of imputation through using the data from the other variables as predictors. The ideal end of the process comes when each variable's, with missing data to start with, distribution converges to a point of insignificant difference between iterations[15].

As stated previously, the method of imputation within the MICE process can be changed, especially for the goal of matching the distribution as closely as possible. Simple methods include imputing each time with the mean of the observed values, although technically this is just an iteration of single imputation. Others make use of the benefit that multiple imputation, and MICE in particular, brings. Within the MICE R function, methods such as predictive mean matching, PMM, bayesian and random forest can be selected. This paper will solely focus on the PMM method but conducting a sensitivity analysis on each of the methods should determine the most useful.

PMM involves a process of regression and nearest neighbour selection to impute its values. Its first iteration takes the observed values to build a regression model, via linear regression, to impute

the missing values. The next step is taking k nearest neighbours. Each imputed data point has k values from the original observed data that are closest to it. The idea is to then take a random value and replace the value from the imputation via regression. This process is repeated in each iteration, where, in this instance, every value including those observed is replaced[9].

- A high percentage of missing data can cause inaccuracies and bias.
- Variables that are not well predicted by the other variables will lead to imputations that are inaccurate.

~~The MICE process is only suitable for data that is not MNAR, some adjustment would be needed to account for this type of data.~~ This paper assumes data is either MAR or MCAR.

2.6 Prediction Models

The primary outcome of interest in this study is the metric evaluating the lung cancer patients' length of stay in the ICU. Therefore, experimenting with machine learning prediction models such as linear regression, penalisation methods and random forests will help identify a suitable and optimal model.

2.6.1 Linear Regression

Linear regression is, compared to other widely known machine learning prediction methods, somewhat simple and easy to understand. It is often used as the first step in many attempts to apply machine learning techniques for prediction purposes. "This paper" shows how a simple linear model can be more effective than other more complex models, with the example of a study involving breast cancer diagnosis prediction[24]. All of which suggests that this method should be evaluated first.

The method itself comes down to the relationship between a dependent variable to be predicted and the independent variables used in the relationship as predictors[18]. Included in this equation is a term as an indicator for the noise evident in the model, in fact one of the main aims in improving the effectiveness of the model is the minimisation of this term. This is referred to as ordinary least squares regression or OLS when the error terms have constant variance.

A univariate version of this model includes only one independent variable for prediction and can be described with a simple equation shown below.

$$y = \beta_0 + \beta_1 x + \epsilon$$

Here, y is the predicted value of the dependent variable. In the case of this report the length of stay in the ICU within a lung cancer patient's admission. β_0 refers to the y intercept given by the equation, which can be visualised by plotting the linear relationship of the model. β_1 is the coefficient of the variable x and is symbolic of the effect the variable has on y. Finally, ϵ represents the error term that indicates the noise in the prediction. ~~The predicted data points can be plotted on a 2D x,y plane along with a line of best fit to see the trend.~~

Since this prediction model is likely to include several independent variables, it would be more apt to model using the equation for multivariable linear regression. The equation's structure is very similar to the equation for univariate regression but with the added terms to represent the effect of the extra variables on, in this case, the lengths of stay. The equation is as follows.

$$y = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p + \epsilon$$

~~The difference becomes clear upon observation.~~ The number of terms in multivariate regression is represented as the letter p in the equation. ~~A more succinct way to express this equation is to write it in matrix form [27] as follows.~~

$$\cancel{Y = X\beta + \epsilon}$$

~~Here, both Y and ϵ are column vectors with dimensions n x 1, X is a matrix with dimensions n x p, and β is a column vector p x 1. The vector Y is a vector of the predicted values, the vector β a vector of the coefficients, and the contents of the matrix X are the data of the variables used in the prediction. The first column in X are all 1's to include the value of the intercept β_0.~~

Since the key goal in improving a linear regression model is minimising the error term the equation that the computer code seeks uses to do this is a rearranged version of the earlier equation which is written to make predictions for the depedent variable y. The rearranged equation simply puts the error term in the place of y. Since the sum of the residuals, a name for the error term values, is always zero minimising the sum of the residuals would not make any sense. Therefore the sum of the squared residuals must be observed to evaluate the model. The equation is called

the residual sum of squares and is shown below.

$$RSS = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2$$

With i representing an individual record summing to the sample size of n and j representing a predictor variable among p predictor variables. So the function to build the model within the code would find the β values to minimise this equation, these β values can then be used in the original equation to predict the dependent y variable.

2.6.2 One Hot Encoding

Categorical data are not handled well by machine learning models because they are designed to handle numerical data. This is where one hot encoding comes in. The idea, to put it briefly, is to assign binary values as an indication as to which category the record falls into. For example if a categorical variable has four categories then four separate columns would be created. Let's say a category in that variable is urgent admission then every admission record included in this category would receive a value of 1 and the others a value of 0. As would be the same for the other categories in that variable

2.6.3 ~~LASSO~~ and Ridge Regression

LASSO and ridge regression models introduce a β coefficient penalisation term which minimises their value towards zero to reduce overfitting in the model. When the model is overfitting it is trained too well to make predictions on the training data. This is a problem because any new data, ~~which in some way would definitely differ from the data the model is trained on in it's composition~~, used to make predictions would produce predictions that may be far from equivalent to the actual values. ~~So~~ these two regression models both introduce a ~~lambda~~ term which needs to be tuned carefully to find ~~the best~~ model that reduces overfitting, improves prediction and does this without creating an overly complex model. ~~So~~ the equation for these two models is very similar to the equation for the OLS. For ridge regression the equation is as follows.

$$RSS + Penalty = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

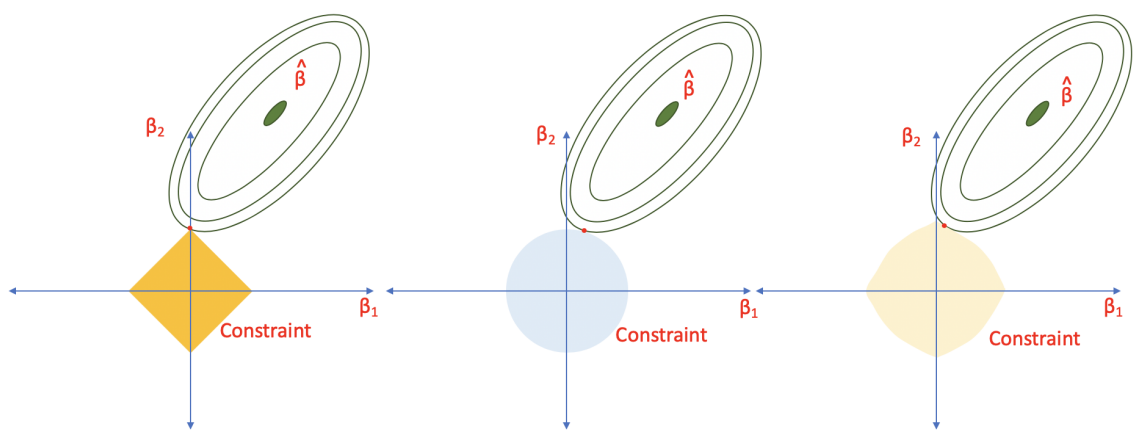
And the LASSO equation is written below.

$$RSS + Penalty = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p}|\beta|$$

In both these equations $\lambda \in [0, \infty)$.

Here we see the difference in the two models and how they are both linked to OLS. Essentially when the value of λ is zero there is no penality so you are left with the minimisation of the RSS alone and an equation for the OLS with the β values being the same as they are in this model.

Plotting the contours of the error and the constraint regions [datasklr cite] help to visualise how the β coefficients are pushed to zero with these methods.



So what can be seen here from the three penalisation model graphs are three differing shapes for the constraint regions. The ellipses represent the contours of the RSS values away from the OLS. The constraint regions will increase in size as $\lambda \to 0$ which would bring the β values to their OLS values. However, if λ does not equal zero then the constraint shrinks along with the β values as they are evaluated at the point where the RSS + penalty contours intersect with the constraint region. The reason why LASSO pushes some coefficients to zero is shown in these plots. Because of its diamond shape there is a possibility that the contours intersect with the constraint region on the axes. This is not possible with ridge regression since in OLS no coefficients have zero values and therefore the contours will not intersect the circular shape at the axes. The third plot from the right shows the constraint region for an elastic net which is the middle ground between LASSO and ridge regression. Here the coefficients may get pushed quite close to zero but will never reach it.

The plots show only a relationship between a dependent variable and two independent vari-

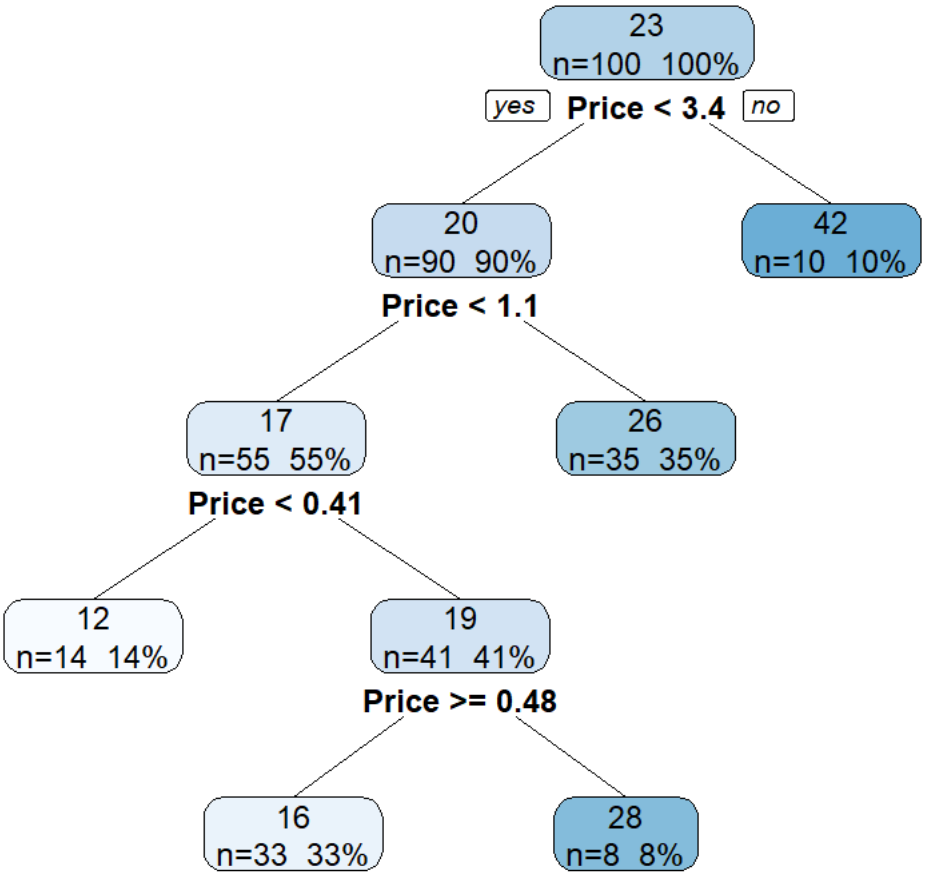
ables, hence the ability to plot this on an x-y plane. For more than two independent variables the plots get more complex and include further dimensions so they are very tricky to display on paper

2.6.4 Random Forest

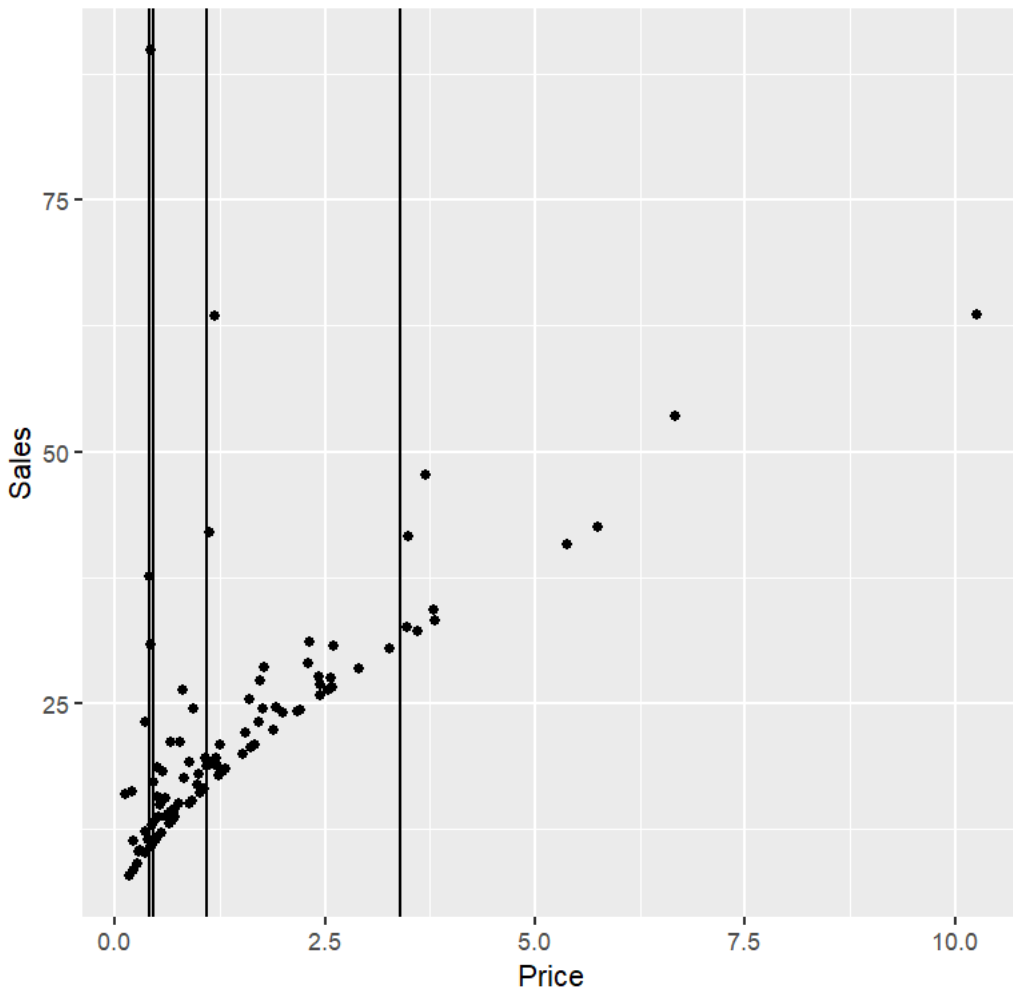
To explain the random forest method, the concept of a decision tree must be established first[25]. At the beginning of constructing a decision tree 100% of the data constitutes what is labelled the root node[23]. From there this node is split in two and then two again and so on. How these splits are determined is based on attributes of the indepent variables used in the model. When training a decision tree the independent variables are used to determine how to split the data into the two new nodes, or leaves. Should a leaf be split in a way that minimises the mean squared error by a significant it will carry on splitting. Given this standard, a leaf that will not be split further is labelled a terminal node. Each terminal node includes a subset of the overall population of the dataset, and each individual is assigned a prediction of the dependent variable which is equivalent to the mean of the dependent variable values in the training set[31]. Therefore when evaluating the model the test data would be fed into the tree and the predictions obtained depend on which leaf each individual falls into. The evaluation metric can then be calculated using these predictions and the actual values.

How the algorithm splits the data is determined by how much a variable or multiple variables minimise the mean squared error after the split[17] and the same is true for the decision on which leaf to split further. In CART trees, classification and regression trees, only one variable is selected[11] and these are the type of trees that are used as the makeup for the ensemble random forest model. All univariate decision trees, by nature, follow the CART model in that only one variable is used to determine each split. However there are other multivariable models that use a linear composition of the variables to consider to determine the splits [6], tree algorithms such as oblique decision trees separate data in this way [12], but these are not used in the random forest R function used in this report.

As an example consider a univariate model where the price of a product, in a non-specific currency, is used to predict how many products are sold. The data consists of 100 products each of different prices and the total sales of each. The trained decision tree for this data is plotted as follows.



From this figure you can see the cascading steps from node to node as the algorithm determines the prediction of the target variable. The value for which each split is decided upon is as described earlier, the algorithm will find the one split that has the most significant minimisation of the mean squared error. Since this example includes a relatively small sample population of data so there is bound to be some unusal predictions. Case in point the price values between 0.41 and 0.48 giving a sales prediction of 28, deviating strongly from the pattern of the association. This comes as a limitation of the single decision tree model, specifically one that has a small sample with some outliers, as it is prone to overfitting therefore not being so useful to predict unseen data.



This is a graph visualisation of the data for the price of the product the products sales with lines that split the graph into sections according to the output of the decision tree model. Upon inspection it can be confirmed that the $0.41 < 0.48$ price group does in fact include a very significant outlier and other higher values than would be expected for that section therefore resulting in an unusually increased prediction.

It is worth noting that this data has been randomly generated and in no way models the actual relationship between the price of a product and its subsequent sales figures.

This example and explanation of basic decision trees lays the groundwork for the explanation of random forest prediction models.

Random forest is an ensemble method, one that is an ensemble of multiple decision trees[10]. This model applies a bootstrapping aggregation, bagging, technique to sample with replacement from the training dataset to create D new datasets that will be used to train D trees[1], where the number of trees can be tuned as a parameter of the model. With random forest the number

of variables randomly selected to consider for each split is limited by the hyperparameter mtry, a value that will always be less than the number of predictor variables. A larger mtry value will increase the amount of bias whereas the a low mtry value will increase the amount of variance, so finding the optimal value is key to find a middle ground in balancing the variance and the bias. This hyperparameter is very influential as it is what separates the random forest out from other ensemble decision tree models. The D trees therefore constitue the forest part of the algorithm each giving separate predictions for the data. The next step of the algorthim is to aggregate the predictions from each of the trees. Since this is done with a k-cross validation these steps are repeated k times. Then the model can be evaluated according to the process of using the test dataset to test the model and produce the values for the performance metrics.

To tune the hyperparameters, such as number of trees and mtry, it is useful to visualise their effect on the variance as their value increases. Then it would be much easier to determine the optimal values of the parameters, those that allow for an effective model without introducing too much computational intensity.

2.7 ~~Validation Methods~~

It is important to consider the validity of prediction models. There are several methods that are designed to check the generalisability of how well models fit to new data which can be assessed both using either internal or external validation[13]. A ~~regression~~ model needs to go through this process of validation so that it can be compared with other models and also it assists in determining whether the model may be suitable for use in practice.

A common method of internal validation is k-fold cross-validation[7]. This involves splitting the dataset into k folds, usually k is set to 5 or 10, one split is used as the test dataset and the others used to train the model. ~~Once the model is trained it can be evaluated on the test dataset. This is then iterated so that each fold is used as the test dataset, meaning that~~ all the data is used both to train and test. The performance metrics in each fold can be evaluated using the predictions and actual values for the test set. ~~Averaging the metrics gives an overall evaluation of the performance of the model.~~ k-fold cross-validation is an improvement on the simpler method of split-sampling, which is essentially a cruder version of the method. Split-sampling could be thought of as equivalent to a 1-fold cross-validation just without rotation on different folds as the data is split into only one training set and one test set. The reduced use of the complete data to both train and test, as is the case in cross validation, introduces a higher level of variability and bias in estimating

the performance of the model[3]. This is why the cross validation technique is highly favoured in comparison.

Leave one out cross-validation is a special case of the k-fold cross-validation method. In this case if there are n records in the dataset then there would be n folds. Effectively in each iteration there is a training dataset of size n-1 and a test set of 1. How this method is performed and evaluated is identical to the k-fold, where k is n. The main advantage of this method over cross validation for a smaller k, is that since in each iteration the model is trained on almost the complete dataset, n-1, so there is a very low bias. Some significant disadvantages, however, is a high variance in the estimator, and a further instability due to outliers in the data or model sensitvity. This is stated quite succintly by Hastie et al, with ”k = n, the cross-validation estimator is approximately unbiased for the true (expected) prediction error, but can have high variance because the n “training sets” are so similar to one another” [31]. Quite often the computational cost disadvantage of running n iterations of training and evaluating the data, is enough to find the leave one out method as unsuitable. Datasets of size beyond 1000 are typically not suited to this method because of this. So all things considered, the leave one out method is less useful in this specific study compared to standard cross-validation.

Bootstrap validation is a sophisticated technique that, since about the 1980s/1990s, has become a viable method of model validation for use on larger datasets due to increased computing power. The bootstrapping method attempts to adjust for the optimism in the performance estimate therefore reducing its bias. Iba et al note that a multivariable prediction model’s ””apparent” predictive performances such as discrimination and calibration calculation in the derivation dataset are better than the actual performance for an external population”[20]. The optimism is then defined as the described difference between the performance on the ”derivation dataset” and the performance for an external populaiton. The process involves creating M new datasets of size N from the original dataset. Each of the new datasets may include multiple of the same records found in the original as they are sampled with replacement [2]. The first iteration involves getting the performance from the original dataset and the performance for the bootstrapped dataset. The optimism of each model can be derived from these two performance values by subtracting the performance on the bootstrapped dataset from the performance on the full dataset. The optimism is averaged over the M bootstrapped datasets to give an average optimism value. Subtracting this value from the full dataset’s performance estimation gives the value for optimism-adjusted performance estimation. This bias in the new performance estimation is reduced because the size of the dataset

used to train the model is equivalent to the original dataset as opposed to the shrinking of the cross validation techniques. Similar to leave one out cross-validation bootstrapping in comparison to k-fold cross-validation is better when the dataset is smaller due to the increased computational complexity and decreased effectiveness in larger datasets.

The validation method used in this study will be k-fold cross-validation due to its managable computational intensity and its ability to work well with larger datasets.

The best way to validate a machine learning prediction model is to validate externally. A model validated only on the data included in the original dataset still carries a lot of optimism and may fit poorly. Evaluation on relevant data obtained in new settings gives a better estimate of how generalisable the model is to a diverse range of populations. The more external validations performed with a more varied array of data provide a greater picture of the generalisability. This is important because if the model is applied to populations with demographics dissimilar to those represented in the training or testing data, the likelihood of inaccurate predictions increases. Inaccurate predictions lead to inefficiencies that reduce the model's usefulness.

2.7.1 Nested Cross Validation

A simple k-fold cross-validation is prone to some bias. A way to reduce this bias is to use a process of cross-validation within each fold, which is a method commonly known as nested cross-validation. A visualisation of the process is given in figure 8 of this paper[22], it shows how the train dataset is split for cross-validation. The bias that arises in standard cross-validation stems from the fact that the test data used in evaluation will also be used in hyperparameter tuning. Since this is the case, a certain amount of data leakage occurs, as the validation test set is essentially being peeked at during training. The leakage leads to an over optimistic assesment of performance, which is the cause of the bias. The nested approach minimises the bias significantly because the hyperparameters are tuned solely on the train datasets. These points echo the understanding that, Krstajic et al [8], present. Of course the downside of nested cross-validation is the increased rounds of training and testing.

2.7.2 Cross Validation to Imputation Pipeline

To properly assess the performance of a model that is trained with data that has missing values needs to go through both a cross validation and imputation step. It is important to consider the

details of the process, mainly deciding the order of the application of the pre-processing imputation and the validation methods. The reason this is so important is due to the need to reduce data leakage[33] in the model. This particular type of leakage is named test-to-train leakage and it describes the idea that if imputation is done before the dataset is split into k-folds there will be leakage because in the process of imputation the test data will be used to affect the imputed data in the training dataset, which is a problem because the test set should be unseen data that has no interference in the training set. So therefore to avoid this type of data leakage it would be advisable to impute the train and test set separately within each cross-validated fold.

When training a given model the amount of seperate models created in this process is equivalent to $K \cdot M$ where K is the number of folds used for cross-validation and M is the number of imputed datasets within each fold. The ultimate aim when training a machine learning model is to have only one model, so the $K \cdot M$ models will have to be aggregated into one. talk about the ~~with() function~~ and what that does and then talk about the pool function and what that does, then about evaluating the aggregated test sets on the pooled model. The performance metrics will be averaged

3 Results

3.1 Dataset Description

3.2 Patient Demographics

Characteristic	N = 2303
Number of previous ICU stays (within hospital admission)	
0	2,115 (92.0%)
1	171 (7.4%)
2	13 (0.6%)
3	3 (0.1%)
4	1 (<0.1%)
Age (years)	
18-40	14 (0.6%)
41-50	93 (4.0%)
51-60	386 (16.8%)

61-70	769 (33.4%)
71-80	668 (29.0%)
81-90	318 (13.8%)
91+	55 (2.4%)
Sex	
Male	1,237 (54%)
Female	1,066 (46%)
Ethnicity	
White	1,671 (72.6%)
Black	217 (9.4%)
Asian	122 (5.3%)
Hispanic/Latino	49 (2.1%)
Other	66 (2.9%)
Missing	178 (7.7%)
Insurance	
Medicare	1,418 (61.6%)
Private	537 (23.3%)
Medicaid	288 (12.5%)
Other	44 (1.9%)
Missing	16 (0.7%)
Admission Type	
Emergency	1,278 (55.5%)
Observation	365 (15.9%)
Urgent	335 (14.5%)
Elective	325 (14.1%)
Platelets (k/μL)	
Mean (SD)	265.8 (133.1)
N missing (%)	1,347 (58.5)
Glucose (mg/dL)	
Mean (SD)	137.2 (78.9)
N missing (%)	1,252 (54.4)
Chloride (mEq/L)	
Mean (SD)	100.4 (8.0)

N missing (%)	1,259 (54.7)
Potassium (mEq/L)	
Mean (SD)	4.7 (4.9)
N missing	1,244 (54.0)
Haemoglobin (g/dL)	
Mean (SD)	10.6 (2.6)
N missing (%)	1,258 (54.6)
Haematocrit (%)	
Mean (SD)	32.7 (6.7)
N missing (%)	1,249 (54.2)
Sodium (mEq/L)	
Mean (SD)	135.3 (13.0)
N missing (%)	1,250 (54.3)
Drugs taken	
Ondansetron	397 (17.2%)
Calcium Gluconate	926 (40.2%)
Death in ICU	
Survived	1775 (77.1%)
Deceased	528 (22.9%)
Length of stay (days)	
Mean (SD)	3.9 (5.2)

Table 1: Caption

3.3 Missing Data

3.4 Imputation

3.4.1 Trace Plots

3.4.2 Correlation Plot

3.4.3 Categorical Missing Values

3.5 Prediction Models

3.5.1 Linear Regression

3.5.2 Penalisation Models

3.5.3 Random Forest

3.6 Model Performance

4 Discussion

5 Conclusion

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Appendix